



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

SAP ABAP-Based Healthcare Data Management and Reporting System

The domain of the Project:

Healthcare Information Systems in ERP

COURSE NAME:

SAP ABAP

Team Mentor:

Name: Santoshkumar Ramlakhan Pal

Designation: SAP ABAP Associate Consultant at Infosys Limited

Team Members:

Ms. Pranowmya Nalam

Period of the project:

November 2025 to April 2026



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Declaration

The project titled “**SAP ABAP-Based Healthcare Data Management and Reporting System**” has been mentored by “**Santoshkumar Ramlakhan Pal**”, organized by SURE Trust, from November 2025 to April 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members: Ms. Pranowmya Nalam

Mentor Name: Santoshkumar Ramlakhan Pal

Designation: SAP ABAP Associate Consultant at Infosys Limited

Prof. Radhakumari

Executive Director & Founder

SURE Trust



Table of Contents

1. Executive summary
2. Introduction
3. Project Objectives
4. Methodology & Results
5. Social / Industry relevance of the project
6. Learning & Reflection
1. Future Scope & Conclusion



Executive Summary

The **SAP ABAP-Based Healthcare Data Management and Reporting System** is designed to demonstrate how enterprise resource planning (ERP) systems can be effectively utilized in the healthcare domain for efficient data handling and reporting.

This project focuses on developing customized SAP ABAP solutions including SmartForms for structured document generation, ALV (ABAP List Viewer) reports for interactive data display, and automation of standard SAP reports using the SUBMIT statement. The system is capable of extracting data from SAP standard tables such as BUT000 and KNA1 and presenting it in a meaningful and user-friendly format.

The key objective of the project is to reduce manual effort, improve data accuracy, and enhance reporting efficiency in healthcare-related processes. By integrating multiple SAP tools and techniques, the project showcases how business data can be transformed into actionable insights.

Key outcomes include

- Development of structured SmartForm outputs.
- Creation of interactive ALV reports with Excel download functionality.
- Automation of report execution using standard SAP programs.
- Improved accessibility and usability of healthcare-related data



Introduction

Background and Context

The healthcare industry generates a large volume of data related to patients, business partners, services, and transactions. Efficient management of this data is essential for ensuring smooth operations, accurate reporting, and informed decision-making.

SAP ERP systems play a crucial role in managing enterprise-level data across various industries, including healthcare. SAP ABAP (Advanced Business Application Programming) allows developers to create customized applications, reports, and forms tailored to specific business needs.

This project leverages SAP ABAP to design a healthcare-focused reporting system that enhances data visibility and automates reporting processes.

Problem Statement

Healthcare organizations often face several challenges in data management, including:

- Inefficient handling of large datasets
- Lack of structured and automated reporting systems
- High dependency on manual processes leading to errors
- Difficulty in exporting and sharing data for analysis



These challenges highlight the need for an efficient, automated, and user-friendly reporting solution.

Scope

The scope of this project includes:

- Development of SmartForms using SAP standard tables
- Creation of ALV reports for interactive data visualization
- Implementation of Excel download functionality
- Automation of report execution using the SUBMIT statement

Limitations

- The project uses standard SAP tables (BUT000, KNA1) .
- No integration with real-time hospital systems.
- Limited to reporting and form generation functionalities.

Innovation Component

The project introduces the following innovative aspects:

- Integration of SmartForms and ALV reporting in a single system
- Excel download feature for enhanced usability
- Reuse of standard SAP reports using SUBMIT
- Simplified and user-friendly reporting interface



Project Objectives

Project Objectives

- To develop SmartForms for structured healthcare data output
- To create ALV reports with interactive features
- To implement Excel download functionality for reports
- To automate report execution using SUBMIT
- To enhance efficiency in healthcare data management

Expected Outcomes

- Fully functional SmartForm output
- Interactive ALV report with user-friendly interface
- Successful implementation of Excel download feature
- Automated execution of standard SAP reports
- Improved reporting accuracy and efficiency



Methodology and Results

Methods / Technology Used

- SAP ABAP Programming
- SmartForms for document generation
- ALV (ABAP List Viewer) for reporting
- Modular programming techniques

Tools / Software Used

- SAP GUI
- SE38 (ABAP Editor)
- SE11 (Data Dictionary)
- SE80 (Object Navigator)
- SMARTFORMS

Data Collection Approach

Data is extracted from standard SAP tables:



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

- **BUT000** – Business Partner data
- **KNA1** – Customer Master data

This data is processed using ABAP logic and displayed using SmartForms and ALV reports.

Project Architecture

The project follows a structured architecture:

1. **Data Extraction Layer** – Retrieves data from SAP tables
2. **Processing Layer** – Applies business logic using ABAP
3. **Presentation Layer** – Displays data via:
 - a SmartForms
 - b ALV Reports
4. **Output Layer** – Excel download and report execution

Key Features Developed

1. SmartForm Development

- Designed structured output format
- Included header, main, and footer sections
- Displayed healthcare-related data clearly

2. ALV Report with Excel Download

- Interactive grid display
- Field catalog and layout customization
- Download option for Excel



3. ME2M Report using SUBMIT

- Called standard SAP report program
- Passed parameters dynamically

Output

The image shows a SmartForm design for an invoice. It includes a header section with fields for 'COPY', 'INVOICE', and 'LOGO'. Below this is a 'BP_DETAILS' section. The main body is labeled 'MAIN' and contains a 'SIGNATURE' field. The footer section includes a 'PAGE' field. The design is displayed on a grid with dimensions indicated on the left and top.

Fig1: SmartForm Design

The image shows the output of the SmartForm design, which is an invoice. It includes a header section with fields for 'ORIGINAL INVOICE', 'INVOICE', and a logo. Below this is a 'Business Partner' section with fields for 'Business Partner', 'Partner Type', 'Business Group', 'Name', and 'Created Date'. The main body contains a table with columns for 'SALES DOC NO', 'CREATED BY', and 'SOLD TO'. Below the table is a table with columns for 'SL.NO', 'ITEMS', 'MATERIALS', 'QUANTITY', and 'AMOUNT'. The footer section includes a 'SIGNATURE' field.

Fig2: Smartforms Output

The image shows the output of an ALV report. It is a table with columns for 'Material Doc', 'Year', 'Posting Date', 'Material', and 'Plant'. The data is displayed in a grid format with a toolbar at the top.

Material Doc	Year	Posting Date	Material	Plant
6300000040	2023	01.06.2023	FLOUR-01	CHI1
6300000041	2023	01.06.2023	FLOUR-01	CHI1
6300000042	2023	25.06.2023	FLOUR-01	CHI1
6300000043	2023	25.06.2023	FLOUR-01	CHI1
6300000100	2023	07.07.2023	00000000000000003110	HY02
6300000100	2023	07.07.2023	00000000000000003087	HY02
6300000110	2023	07.07.2023	00000000000000003110	HY02
6300000110	2023	07.07.2023	00000000000000003087	HY02
6300000120	2023	07.07.2023	00000000000000003110	HY02
6300000120	2023	07.07.2023	00000000000000003087	HY02
6300000121	2023	07.07.2023	00000000000000003110	HY02
6300000121	2023	07.07.2023	00000000000000003087	HY02
6300000122	2023	07.07.2023	00000000000000003110	HY02
6300000122	2023	07.07.2023	00000000000000003087	HY02
6300000123	2023	07.07.2023	00000000000000003110	HY02
6300000123	2023	07.07.2023	00000000000000003087	HY02
6300000124	2023	07.07.2023	00000000000000003110	HY02
6300000124	2023	07.07.2023	00000000000000003087	HY02

Fig3: ALV Report Output

The image shows the output of an Excel download report. It is a table with columns for 'Material', 'Year', 'Posting Date', 'Material', 'Plant', 'Quantity', and 'Amount'. The data is displayed in a grid format.

Material	Year	Posting Date	Material	Plant	Quantity	Amount
6.3E+09	2023	01.06.2023	FLOUR-01	CHI1	1	0
6.3E+09	2023	01.06.2023	FLOUR-01	CHI1	3	0
6.3E+09	2023	25.06.2023	FLOUR-01	CHI1	4	0
6.3E+09	2023	25.06.2023	FLOUR-01	CHI1	2	0
6.3E+09	2023	07.07.2023	3110	HY02	10	0
6.3E+09	2023	07.07.2023	3087	HY02	10	0
6.3E+09	2023	07.07.2023	3110	HY02	5	0
6.3E+09	2023	07.07.2023	3087	HY02	6	0
6.3E+09	2023	07.07.2023	3110	HY02	10	0
6.3E+09	2023	07.07.2023	3087	HY02	5	0
6.3E+09	2023	07.07.2023	3110	HY02	5	0
6.3E+09	2023	07.07.2023	3087	HY02	10	0
6.3E+09	2023	07.07.2023	3110	HY02	7	0
6.3E+09	2023	07.07.2023	3087	HY02	3	0
6.3E+09	2023	07.07.2023	3110	HY02	5	0
6.3E+09	2023	07.07.2023	3087	HY02	10	0
6.3E+09	2023	07.07.2023	3110	HY02	7	0
6.3E+09	2023	07.07.2023	3087	HY02	3	0
5E+09	2023	15.07.2023	FLOUR-TE	3000	100	0
5E+09	2023	15.07.2023	FLOUR-01	CHI1	100	0
6.3E+09	2023	17.07.2023	3110	HY02	5	0
6.3E+09	2023	17.07.2023	3087	HY02	7	0

Fig4: Excel Download Output



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

The image shows the SAP input screen for a contract. It features a dark blue header with the SAP logo and a navigation bar with a checkmark, a dropdown menu, and buttons for 'Save', 'Cancel', and 'Help'. Below the header, there are two input fields: 'P_MATNR' with the value '1' and 'P_WERKS' with the value '1000'. The 'P_WERKS' field has a small icon next to it, possibly indicating a selection or confirmation action.

Fig5: Input Screen

Contract	Type	Vendor	Name	Pg	Agmt	Date
Item	Material	Short Text				Mat. Group
D I A Plnt Sloc		Targ. Qty Un	Net Price	Curr.	per Un	
4600000119 WK 3701	eSupplier, Inc			002	05.03.2013	
00001	Service Item					00103
D U		1 AU	0.00	USD		1 AU
4700000014 DC 50000	ICYOU Ltd.			001	21.08.1995	
00020	Lether					016
X W		0.000	0.00	PTE		1
00030	Lether					016
X M		10.00 M2	0.00	PTE		1 M2
5001000050 GCTR 1000	sri			002	08.12.2003	
00001	Test					001
U		11 PC	0.00	EUR		1 PC
5001000051 GCTR 1000	sri			001	08.12.2003	
00001	test					001
U		11 PC	0.00	EUR		1 PC
5001000053 GCTR 1000	sri			002	09.12.2003	
00001	tttt					001
U		11 PC	0.00	EUR		1 PC

Fig6: Output Screen

Results

- Improved report generation speed
- Reduced manual intervention
- Increased data accuracy
- Enhanced user experience through interactive reports



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Social / Industry Relevance of the Project

This project is highly relevant in the healthcare sector where efficient data management is critical. It can be applied in:

- Hospitals and clinics
- Healthcare administration systems
- Medical data reporting systems

Impact

- Reduces manual workload
- Improves operational efficiency
- Enables data-driven decision-making
- Supports digital transformation in healthcare



Learning and Reflection

Technical Learnings

- SmartForm design and development
- ALV report creation and customization
- Data extraction from SAP tables
- Use of SUBMIT statement
- Debugging and error handling

Management Learnings

- Time management
- Project planning and execution
- Problem-solving approach
- Independent project handling

Overall Experience

Working on this project was a highly valuable learning experience. It provided practical exposure to SAP ABAP development and helped bridge the gap between theoretical knowledge and real-world applications. The project enhanced both technical expertise and confidence in handling ERP-based solutions.



Conclusion and Future Scope

Conclusion

The project titled “**SAP ABAP-Based Healthcare Data Management and Reporting System**” successfully demonstrates the practical application of SAP ABAP in the healthcare domain. The main objective of developing an efficient data management and reporting system has been achieved through the implementation of SmartForms, ALV reports, and automated report execution using the SUBMIT statement.

The SmartForm developed in this project provides a structured and organized way to display healthcare-related data, while the ALV report enhances user interaction by offering a clear and flexible data view along with Excel download functionality. The use of the SUBMIT statement to execute standard SAP reports like ME2M highlights the ability to reuse existing SAP functionalities effectively without redundant development.

Overall, the project improves data accessibility, reduces manual effort, and increases reporting accuracy. It also demonstrates how ERP systems like SAP can be customized to meet domain-specific requirements in healthcare. This project serves as a strong foundation for understanding real-time SAP development and its impact on business processes.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Future Scope

- Integration with SAP Fiori for improved user interface
- Development of real-time dashboards
- Integration with live healthcare systems
- Advanced analytics and reporting features
- Automation using AI and machine learning